MIAMA UI data authority workflow

Draft one-way contract from Tab 2 volume inputs to Tab 5 results

⚠ Design sandbox

This document describes the intended one-way UI contract. It also records where the current implementation still departs from that contract. It should be merged into the main methodology only after the remaining allocation rules are agreed and implemented.

Core principle

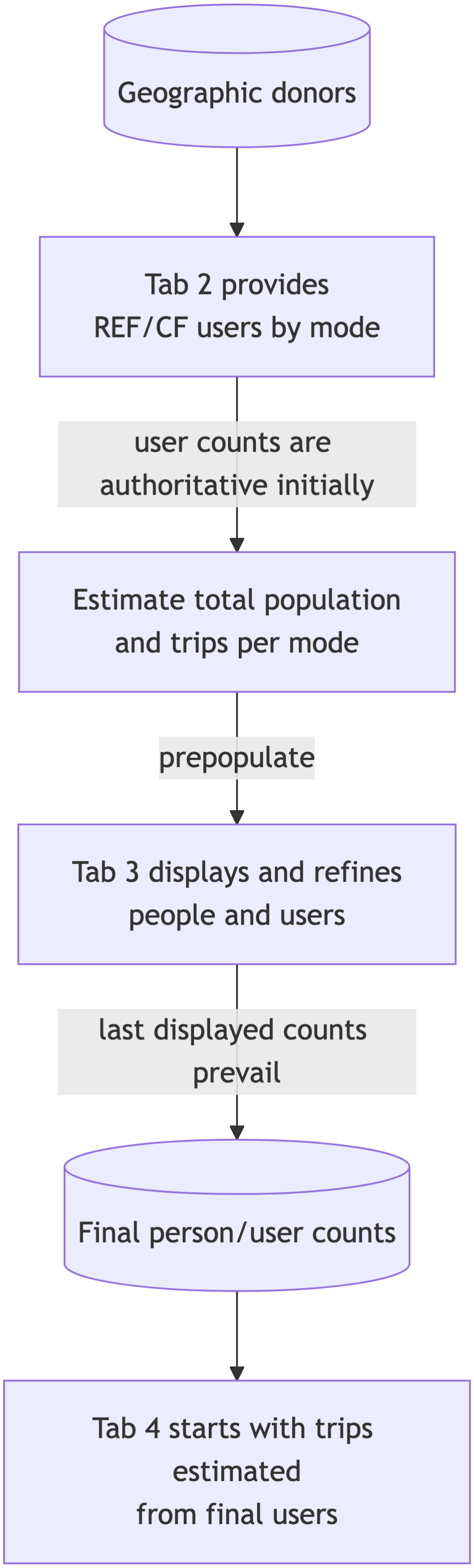
The quantity explicitly provided by the user has authority. Downstream tabs may refine quantities that MIAMA inferred to complete the scenario, but they must not silently feed those estimates back upstream and replace the provided quantity.

1. Tab 2 establishes the authoritative active-travel volume in the selected unit.
2. MIAMA estimates the missing person/trip quantities needed to construct REF and CF snapshots.
3. Tab 3 may revise the person allocation. It does not recalculate an authoritative Tab 2 trip-derived volume.
4. Tab 4 may revise trip counts and characteristics. It does not recalculate the final Tab 3 population or mode-user counts.
5. A later explicit edit supersedes an earlier estimate for that same quantity. It does not retroactively alter a different authoritative unit.

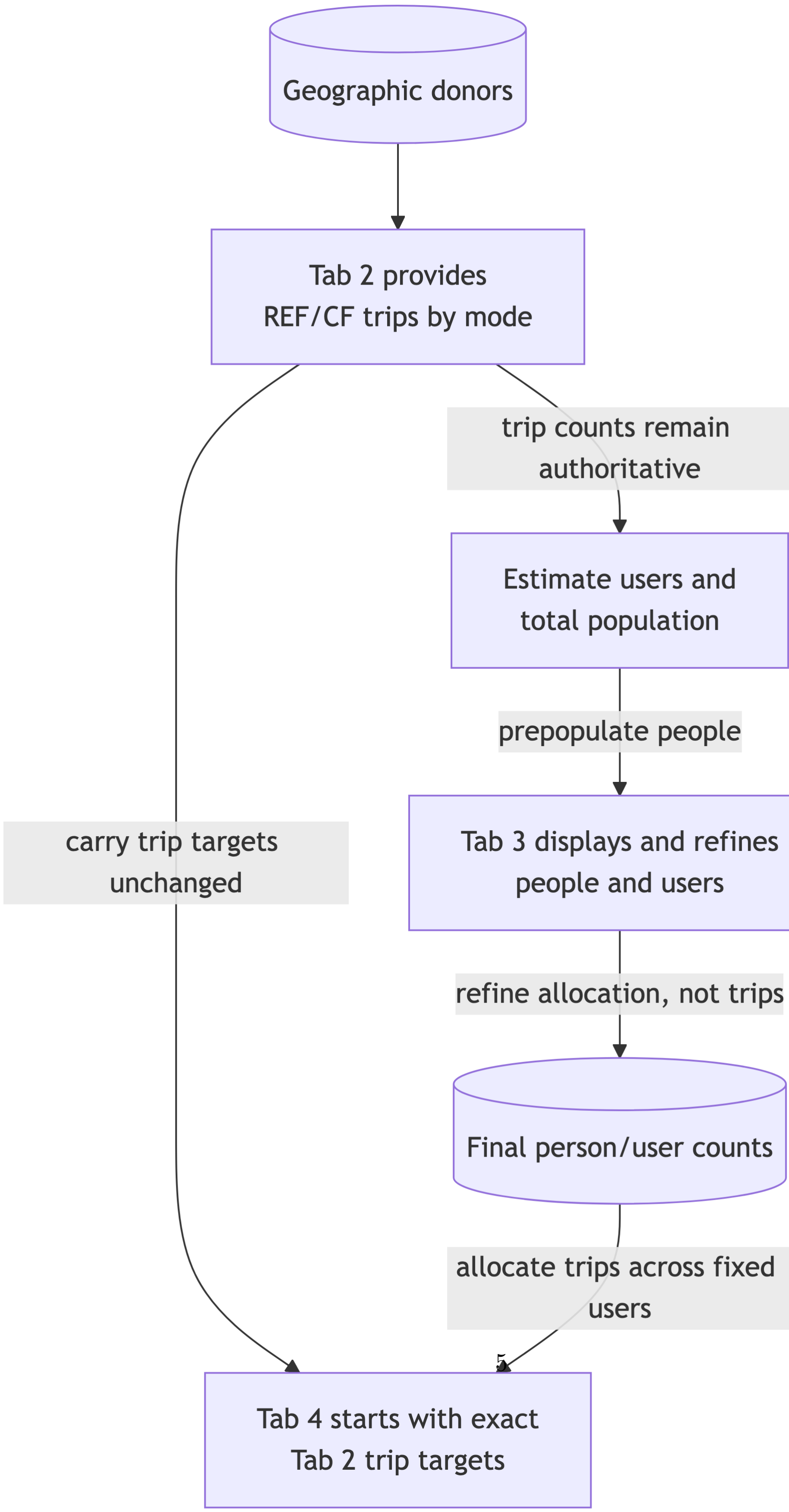
Review-sized workflows

These smaller diagrams contain the same logic as the comprehensive workflow but are intended for normal browser and presentation viewing.

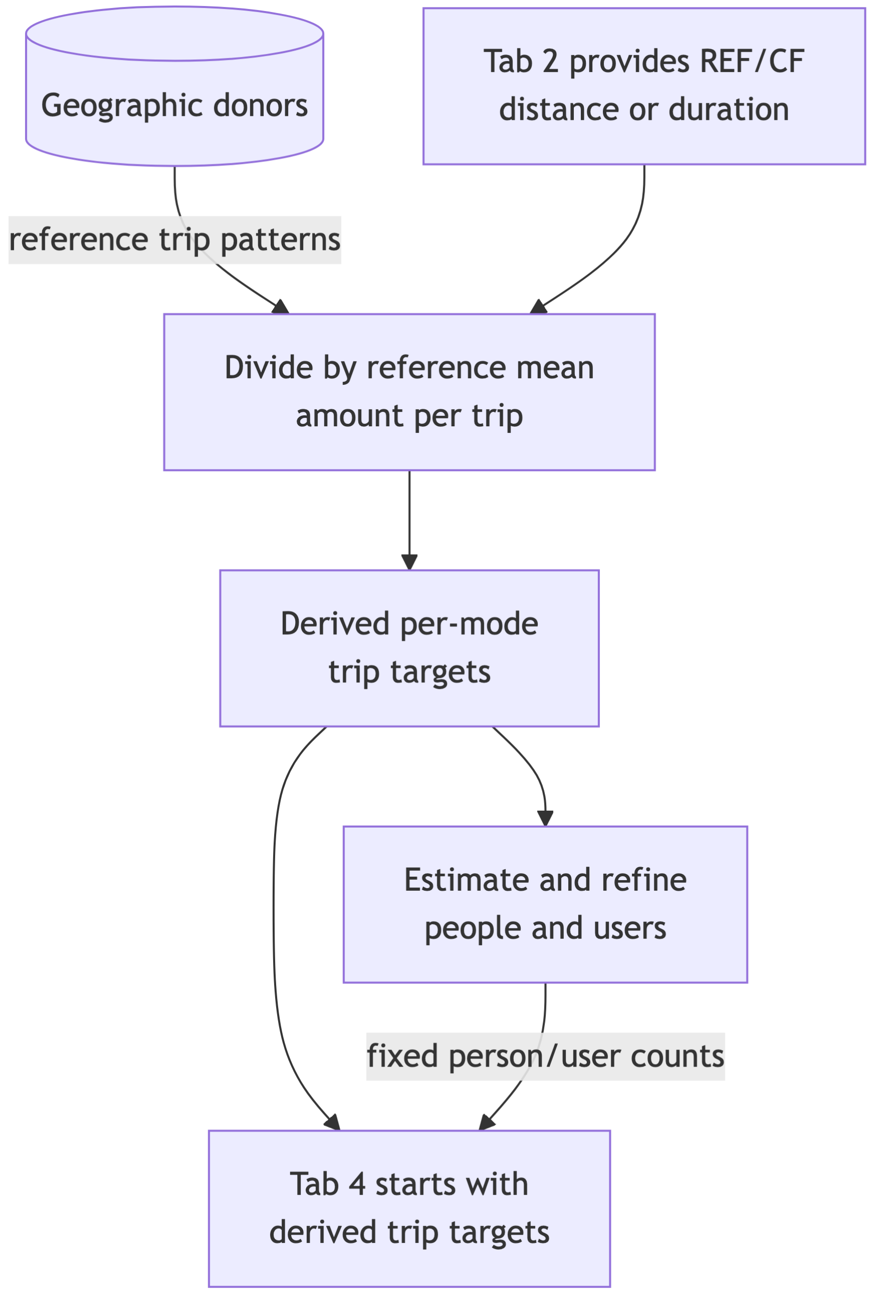
User-count route



Trip-count route

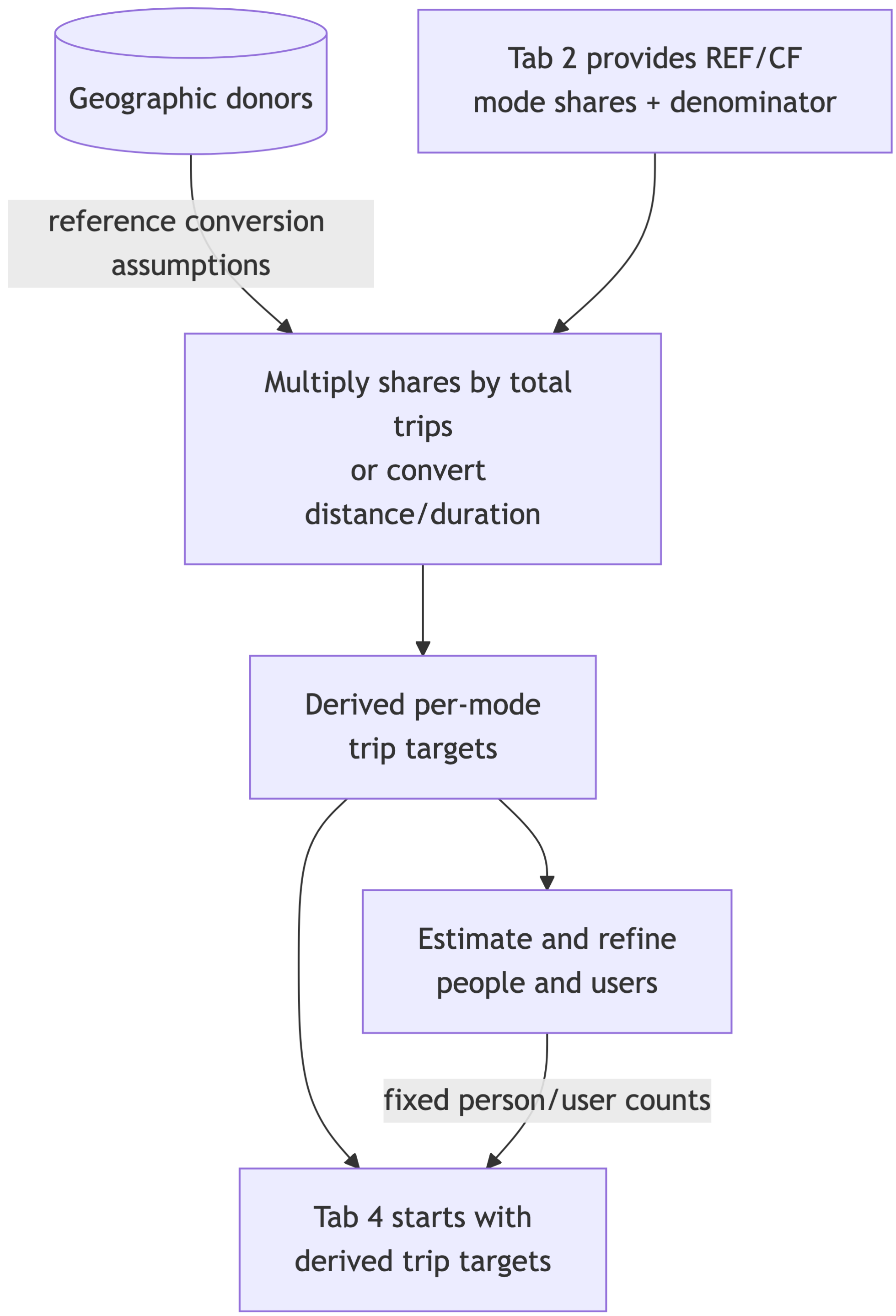


Distance/duration route

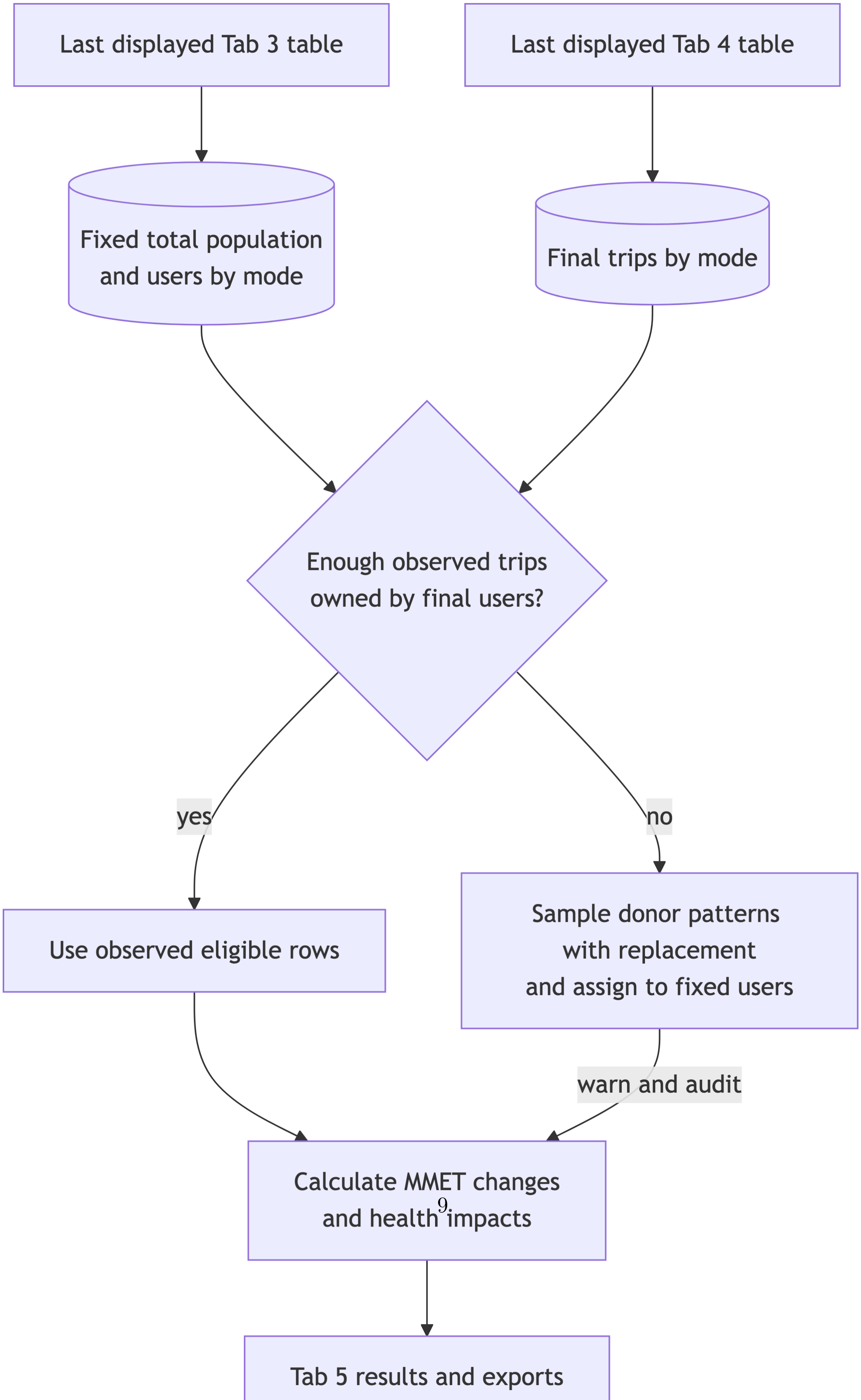


The original kilometres or minutes explain how the starting trip targets were derived. If Tab 4 changes counts or distance assumptions, final aggregate distance/duration is measured from the resulting trip sample.

Mode-share route

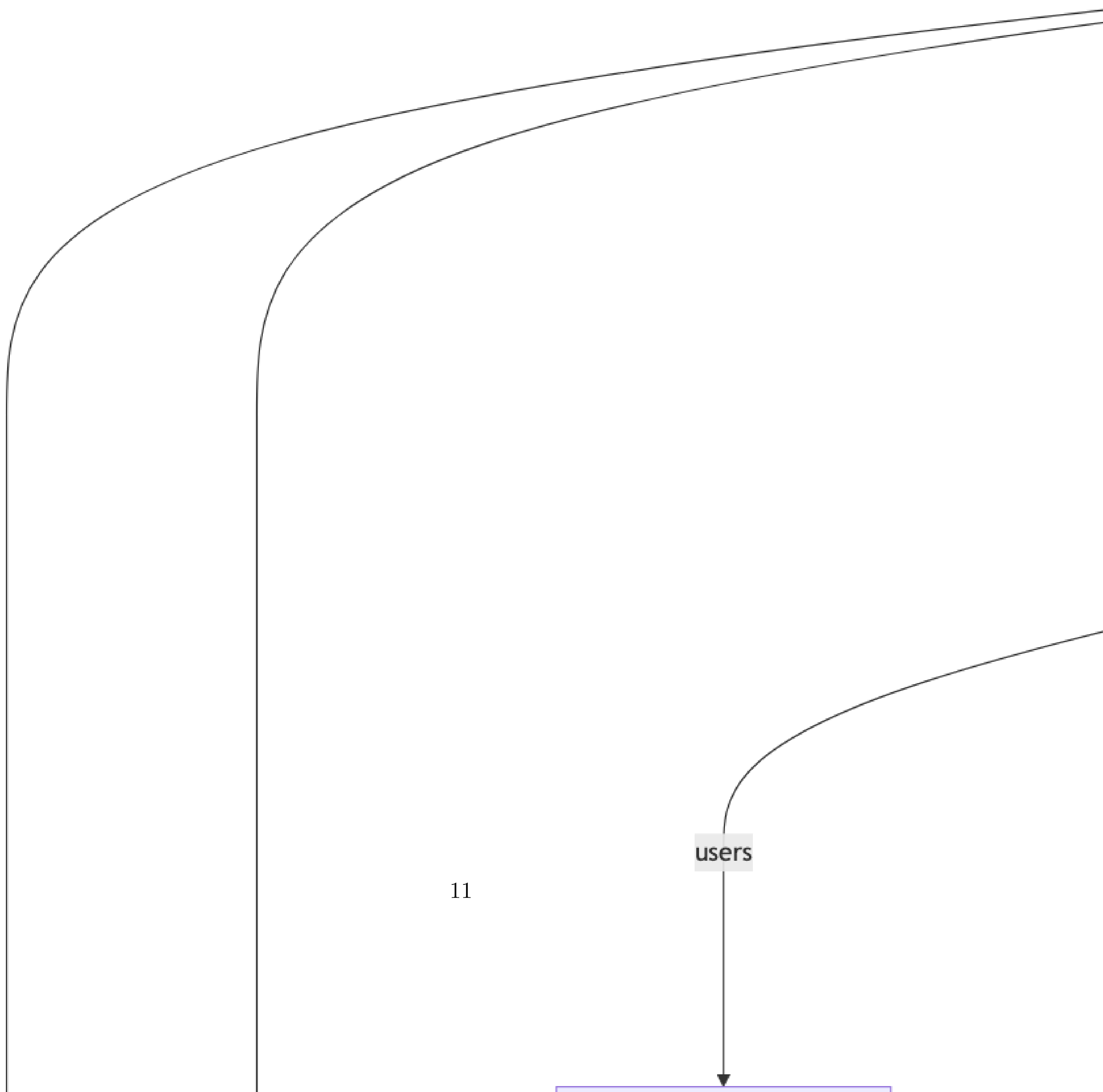


Shared Tab 3-to-results handoff



Comprehensive vertical workflow

The complete diagram is deliberately rendered at full size. Scroll horizontally inside its frame; browser zoom can then be used without making its labels unreadably small.

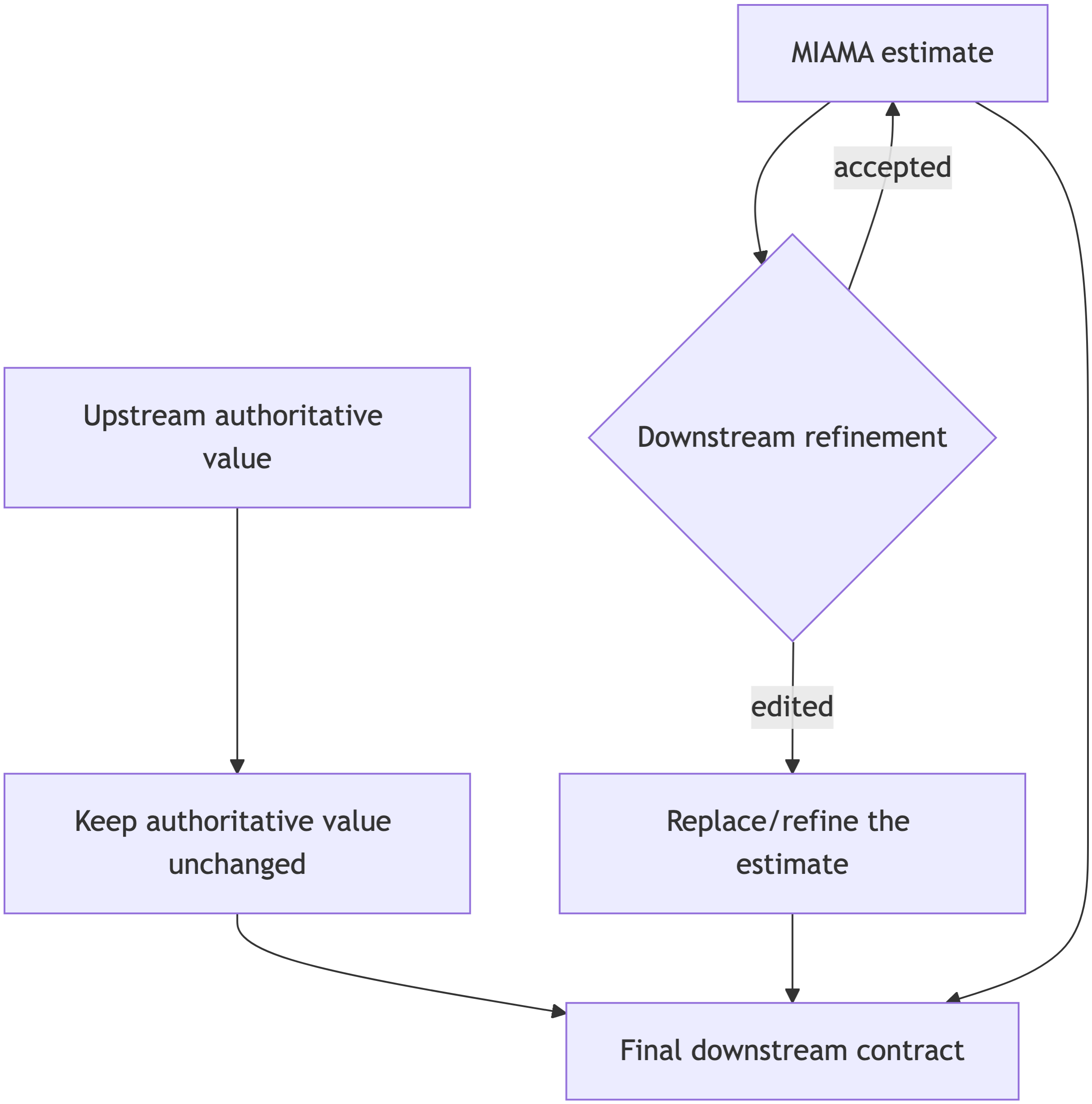

users

Authority by Tab 2 input route

Tab 2 route	Authoritative after Tab 2	MIAMA must estimate	What Tab 3 may change	Tab 4 starting value
Users	REF/CF users by mode	Total population and trips	Total/person allocation and mode-user counts	Trips estimated from the final Tab 3 users
Trips	REF/CF trips by mode	Trip owners, mode users, and total population	Person allocation and user estimates only	Exact Tab 2 trip counts
Distance/duration	REF/CF distance or duration by mode	Trips, users, and total population	Person allocation and user estimates only	Trip counts derived from the Tab 2 volume
Mode share	REF/CF shares and their total denominator	Per-mode trips, users, and total population	Person allocation and user estimates only	Per-mode trip counts derived from shares

An explicit Tab 4 trip-count edit becomes authoritative for trip count from that point onward. It must be allocated across the already-final Tab 3 people; it must not alter total population or mode-user counts.

Refinement effects



Examples:

- Trips entered in Tab 2 remain fixed while Tab 3 changes their allocation across people. Fewer users therefore imply more trips per user.

- Users entered in Tab 2 determine the initial person contract. Tab 3 can revise it, after which trips are re-estimated once for Tab 4.
- Trips edited in Tab 4 are allocated across the final Tab 3 users. More trips therefore imply more trips per user, not more users.

Implementation status

The one-way count contract is implemented as follows:

1. **Trip authority after Tab 3:** Tab 2 trip targets remain present when Tab 3 re-scopes the population. Tab 4 therefore starts from the requested trip totals rather than re-estimating them from whichever rows the selected people happened to own.
2. **Final user contract at results time:** accepted advanced total-population and mode-user counts are reinstated as explicit calculation targets. A Tab 4 trip edit therefore changes trips per user rather than deriving a new user count.
3. **Allocation feasibility:** if the final people own too few observed trips, HUB samples observed donor trip patterns with replacement, assigns them to the fixed mode-user pool, warns, and reports the number of modeled rows in `reference_scope_report$trips`.

The treatment of aggregate units is intentionally different:

4. **Distance/duration and mode-share provenance:** these Tab 2 inputs generate the starting per-mode trip targets. An explicit Tab 4 trip count supersedes those targets. HUB does not force the original aggregate kilometres, minutes, or mode shares afterward; it measures their realized values from the final sampled trips.

Resolved interpretation

1. Donor trip patterns may be sampled with replacement when the final users do not own enough observed rows.
2. Explicit Tab 4 trip counts supersede trip targets derived from Tab 2 mode share, distance, or duration. Remaining-mode shares are measured from the final sample; they are not forced through a second scaling calculation.
3. Final realized distance and duration adjust to the sampled trips and any explicit Tab 4 trip-distance assumptions.
4. The accepted Tab 3 total-population and mode-user numbers are fixed for Tab 4 and results. Tab 4 may change activity assigned per user, not user counts.

Current calculations preserve those counts and use reproducible seeds. Exact person identity across a later full rebuild is not yet serialized as a separate contract object.

The fourth option is the cleanest architectural boundary: finalize a person contract once at the Tab 3-to-Tab 4 transition, finalize a trip contract once at the Tab 4-to-results transition, and make result calculation consume those objects without re-deriving upstream quantities.